

Smart Office case

Day in Data, Tilburg University

Welcome to the innovative world of Capgemini's SmartOffice, where we collect, process, and analyze data to enhance the office experience! At our head office in Utrecht, Leidsche Rijn, we have installed a large network of sensors that continuously collects data on:

- Space occupancy
- Desk occupancy
- Comfort levels, such as CO2, temperature, lighting, and humidity

The data we have gathered can provide valuable insights and predictions and help us improve how we use our Capgemini office. This could mean using meeting rooms more efficiently, improving energy management to reduce costs, promoting sustainability, or implementing predictive maintenance to increase operational efficiency. You are invited to help us with this challenge!

We have chosen three specific topics focused on optimizing our office management.

Case 1: Space Optimization and Capacity Planning for Office Environments

Problem Statement: The use of office spaces fluctuates, with meeting rooms often fully booked while desks and workspaces are underutilized. Additionally, meeting rooms are often under-occupied, for example, rooms for 10 people being reserved by 2 people. This may lead to inefficient use of available office space. Some meeting rooms are used for short meetings or phone calls, which may mean that the available space is not optimally utilized. At the same time, certain departments may be under- or overrepresented in terms of available workspaces.

Goal: Investigate how the current office space can be better utilized by analyzing occupancy data trends and developing a capacity plan. Consider insights such as:

- How meeting rooms and workspaces can be used more efficiently.
- How workspaces and meeting rooms can be better redistributed per department, based on actual occupancy data.

Case 2: Energy Management and Cost Reduction

Problem Statement: Capgemini is continuously optimizing its energy usage. The occupancy of spaces varies greatly throughout the day, week, or month. To ensure a sustainable and efficient working environment, solutions are sought to ensure that meeting rooms, workspaces, and other office spaces are only heated, cooled, or lit when truly necessary. This way, unnecessary energy consumption is limited, and operational costs efficiency is optimized. Additionally, the indoor climate, such as temperature and air quality, remains an important focus to promote employee well-being and productivity and minimize absenteeism.

Goal: The goal of this case is to develop a solution for more efficient energy management within office environments. Consider insights such as:

- Automatic adjustment of temperature and lighting based on occupancy data to reduce energy consumption in unused spaces.
- Predicting energy peaks by analyzing historical occupancy data, so energy management can be aligned with peak moments.
- Reducing the carbon footprint by consuming energy only where and when needed, thereby lowering operational costs and energy waste.

Case 3: Predictive Maintenance for Smart Office Environments

Problem Statement: Maintenance and cleaning in office buildings are often carried out according to a fixed schedule, without considering the actual use of spaces. This could result in inefficiencies, with some spaces being cleaned or maintained unnecessarily often, while other spaces may be overlooked or not addressed in time. Traditional maintenance strategies are reactive or based on static schedules, which can lead to unnecessary costs, waste, and disruptions for building users.

Goal: The goal is to make maintenance and cleaning more dynamic and aligned with the actual use and condition of spaces, installations, and devices. By developing predictive models based on sensor data, maintenance can be better planned, failures can be prevented, and costs can be saved by intervening at the right time. Consider insights such as:

- Predicting when maintenance or cleaning is needed based on actual usage patterns and sensor data.
- Scheduling maintenance more efficiently to minimize downtime or disruptions.
- Extending the lifespan of equipment and installations through timely interventions, such as systems, lighting, or sanitary facilities.
- Realizing cost savings by avoiding unnecessary maintenance or too frequent cleaning and optimizing maintenance patterns.

Final Presentation

For the final presentation of approximately 5 minutes, you are expected to cover the following components:

- What did you investigate?
- What are your recommendations?
- What supports these recommendations?

Variable explanation

building	The name of the building where the points of interest (POIs) are located.
building_id	A unique identifier for the building
floorIndex	The specific floor where the POI is situated.
poiType	Type of sensor, each sensor measures something else: Desk: Occupancy Room Conference: Occupancy, temperature, and humidity, with multiple sensors. Bubble: Similar monitoring capabilities as Room Conference. Comfort: Temperature, humidity, CO2, sound and lux levels. Hidden Room: Not displayed in the app but functions like Room Conference.
poiTypeid	POI ID: The unique identifier for the POI (poiType).
poiName	POI Name: The name used to refer to the POI (poiType).
poiTitleEn	Room number
capacity	Capacity for the room (only visible for room conference)
email	Email of the room
x	The x coordinates of the POI on the floor plan.
y	The y coordinates of the POI on the floor plan.
sensor	A distinct identifier for the sensor monitoring the POI.
sensorCnt	The total number of sensors connected to the POI.
time	The timestamp indicating when the sensor data was recorded.
deviceName	A distinct identifier for the sensor monitoring the Point of Interest (POI).
fCnt	A counter that tracks the number of messages sent by the sensor; resets at 2000 to verify functionality.
occupied	A binary status that changes from 0 (not occupied) to 1 (occupied) based on detected presence.
tempC	The temperature in degrees Celsius measured by the sensor.
humidity	The relative humidity percentage measured by the sensor.
co2	The concentration of carbon dioxide (CO2)
sound	The sound level detected by the sensor, measured in decibels (dB).
lux	The illuminance level measured in lux, indicating the intensity of light.